UTM UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SCSM2113 - HUMAN-COMPUTER INTERACTION

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PROJECT PROPOSAL 2

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Section: 01

Group: Team 3

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PART 1 : USER ANALYSIS

TITLE : GATHERING REQUIREMENTS - USER ANALYSIS

PROPOSED TASKS :

1. **Monitor Equipment or System Information**
   This task involves monitoring real-time data from the equipment and systems to ensure that they are functioning well within the acceptable parameters. To assess the health of the equipment and detect any changes, users need to interpret the sensor readings, status indicators, and performance metrics. Usually, the maintenance engineers and the rig supervisors are the one that are responsible for this task to keep track of machinery performance, while trainees may practice it as part of the learning process.

2. **Locate Issues on Equipment or Around the Rig**
   The task involves identifying or locating faults that arise in equipment or around the rig. The problems should be diagnosed by analysing the system alerts, alarms, or physical symptoms that indicate a fault. In this task, it requires both the ability to assess in real-time data and historical trends, to determine the cause of the fault. Maintenance engineers and rig supervisors typically are the one who will be handling this task, while the trainees may assist under supervision.

3. **Follow Structured Steps and Procedures to Perform Task Safely**
   In this task, the users are required to follow structured procedures when performing the task to make sure of their safety, especially in high-risk environments like oil rigs. This includes adhering to safety protocols, operating procedures, and maintenance guidelines to prevent any accidents or system failures. Every of the workers in the place are supposed to follow these procedures to ensure the safety of both personnel and equipment.

PERSONAS :

PERSONA 1 : Kavinesh Thorne



Figure 1 : Maintenance Engineer

Kavinesh Thorne is a 34 years old senior maintenance engineer in an oil rig company, with a 10 years of experience of working on offshore oil rigs. Most of his 12-hours shifts are spent by inspecting high-pressure pumps, pipelines, and valves under harsh conditions. Kavinesh values safety and efficiency, but he is often frustrated by the need to stop his work and remove his gloves to access the data of the digital tools. This situation makes his workflow become slower.

Kavinesh is motivated by the safety of his crew and the need to complete the inspections in a short time. His goal is to verify the real-time sensor data without interrupting his physical task. He is struggling with the physical interaction that is required for the use of the systems. He need a more efficient solution to be able to access the machine data without leaving the site or even breaking his inspection flow.

PERSONA 2 : Charan Jenkins



Figure 2 : Oil Rig Supervisor

Charan Jenkins is 45 years old and works as a supervisor. She is in charge of the rig's operations and crew safety. With over 20 years of experience, her main task is to monitor the status of the equipment and also respond to the critical alarms. However, sometimes she might face information overload because of the overwhelming amount of data displayed by the multiple screens. This situation makes it harder for her to quickly identify the high-priority alarms. She wants a simplified dashboard that will make it clear for her to make quick decisions in emergencies without wading through irrelevant data.

Charan's goal is to maintain the crew's safety by answering to the critical alarm quickly. She wants a simplified interface that would allow her to make quick decisions when there is an emergency. It will also improve her ability to respond on time to safety issues.

PERSONA 3 : Raghaavesh



Figure 3 : Trainee

Raghaveesh is 23 years old and is currently working as a trainee in an oil rig company. As a newcomer, he is still trying to adjust to the fast-paced and physically demanding work environment. He finds that the existing Digital Twin System is overwhelming since it contains a large amount of technical data and a complex terminology. He is unsure about how to interpret the terror messages and often needs additional guidance to resolve the issues.

To succeed in her role, Raghaveesh hopes to have a step-by-step guide and simplified instructions to help him navigate the system and also perform the routine checks. He wants to rely on visual aids and context-sensitive help to understand the system's diagnostics and error resolution procedures, which will help him build confidence in her role and learn quickly.

SCENARIOS :

PERSONA 1: Maintenance Engineer

SCENARIO 1
Kavinesh is inspecting a high-pressure pump but then he notices an anomaly in the sensor data. However, because of the loud environment and his gloves, he is having a hard time interacting with the screen. After removing his gloves, he walks to the control room, and retrieves the data, which causes delay and he has to leave the equipment unattended

SCENARIO 2
Kavinesh is inspecting a valve then he hears unusual vibrations which trigger an alarm. To assess the fault, he has to check the equipment's data. Due to the noisy environment and his bulky gloves, Kavinesh cannot interact with the system's touchscreen. He removes his gloves and walks to the control room to retrieve the data, which delays his inspection and leaves the valve unmonitored.

SCENARIO 3
Kavinesh is following a step-by-step maintenance procedure for a valve but struggles to interact with the touchscreen due to his gloves. He removes his gloves and walks to the control room to access the procedure, which interrupts his workflow and increases the time needed to complete the task. This delay also leaves the equipment unattended, affecting his overall efficiency.

PERSONA 2: Oil Rig Supervisor

SCENARIO 1
Charan is tasked with monitoring a critical piece of equipment on the rig. He logs into the system and navigates to the appropriate interface to locate the equipment. After finding it, he retrieves the real-time sensor data and begins interpreting the metrics. Upon analyzing the data, he flags the system as normal, as all readings are within safe parameters. Charan then reports the status of the equipment, ensuring his team is updated on the system's condition.

SCENARIO 2
Charan is supervising a maintenance task where the team needs to follow a guided procedure. He first retrieves the maintenance guide by accessing the system and searching for the required task.

After confirming the safety precautions, Charan ensures that the team is ready to begin the maintenance procedure. They execute the steps, checking each part of the equipment, performing actions as required, and marking the task as complete once all steps are finished. Charan monitors the process to ensure everything is done according to safety standards.

SCENARIO 3

Charan is conducting a practice simulation for the team to improve their response to potential emergency situations. She initializes the simulation by toggling the mode and selecting the relevant drill scenario. The simulation runs, and Charan observes the team's performance while they respond to the scenario. After the simulation is complete, she reviews the score and feedback, reading the detailed feedback to identify areas for improvement. If the score is low, Charan ensures the team understands the areas they need to focus on and encourages them to retry the simulation to enhance their skills.

PERSONA 3 : Trainee

SCENARIO 1

Raghaavesh is starting his shift and needs to monitor the rig's status. He logs into the control system, adjusts the display filters, and checks the key performance indicators (KPIs) to ensure everything is operating within normal ranges. Upon reviewing the alarms, he notices an anomaly. He acknowledges the alarm and investigates the issue further by opening the detailed screen for the affected system. After interpreting the data, he decides that immediate action is required and informs the relevant personnel while recording the findings.

SCENARIO 2

Raghaavesh receives a report from the crew about an abnormal system reading. He observes the situation through CCTV and verifies the problem by checking the system information for related parameters. Upon confirming the issue, Raghaavesh narrows down the likely cause and identifies the affected subsystem. He proceeds to arrange a physical inspection, informs the crew, and confirms that the area is safe for them to inspect. After assessing the fault, he escalates the issue to the appropriate personnel and logs the fault in the system for further action.

SCENARIO 3

Raghaavesh is tasked with following a structured procedure to fix a malfunctioning system. He receives the task, reviews the scope, and identifies the correct procedure. Raghaavesh then prepares for the job by reviewing the necessary safety protocols and conducting a pre-job safety process. After ensuring that all safety measures are in place, he arranges for a physical inspection. While performing the task, Raghaavesh continuously observes the crew's behavior and system indicators. He completes the procedure, signs off on the job as completed, and ensures that the system is restored to normal operating conditions before reporting the completion.

PART 2 : TASK ANALYSIS
TITLE : GATHERING REQUIREMENTS - TASK ANALYSIS

INTRODUCTION

During the Task Analysis phase, the focus is on understanding the specific tasks users need to perform within the Onboard Tracker system and how they will interact with it. By interviewing the users, we aim to identify key challenges and inefficiencies that they might face while completing these tasks. This will enable us to design an interface for the Onboard Tracker that is intuitive, user-friendly, and tailored to the needs of the users. The findings from this analysis will directly inform the design requirements of the system, ensuring it meets the needs of all user groups effectively and enhances their experience when interacting with the Onboard Tracker.

HIERARCHICAL TASK ANALYSIS

Based on Persona 1, who is a maintenance engineer in oil rig company :

HIERARCHICAL TASK 1 Plan: Monitor System

0.Monitor System

- 1. Prepare to monitor
  - 1.1 Put on safety gear
  - 1.2 Select system to monitor
- 2 Retrieve real-time data
  - 2.1 Display sensor readings
  - 2.2 Access schematics overlaid on equipment
  - 2.3 Apply filters to highlight critical parameters
- 3 Interpret data in real time
- 4 Take corrective action
  - 4.1 minor, confirm action in log book, continue monitoring
  - 4.2 moderate - follow repair steps,notify supervisor
  - 4.3 critical - follow emergency steps to isolate the system
- 5 Record and communicate
  - 5.1 Log sensor readings, anomaly detection, and action taken on equipment
  - 5.2 Notify supervisor or team if critical
  - 5.3 Continue monitoring equipment or select next equipment.

Plan 1: If not functioning correctly,1-2-3, else 2-4-5
Plan 3: If sensor reading normal, 3-5, else 3-4-5
Plan 4: If minor, follow 4.2,if moderate follow 4.2, if critical-follow 4.3 then 5.2

Video on Explanation of HTA:
https://youtu.be/0MpKTmTERVk
HIERARCHICAL TASK 2 Plan: Identify Faults on Equipment
0 Identify faults equipment
    1 Examine equipment
    2 Detect anomalies
    3 Verify location and severity of issue
    4 Take corrective or investigative action
    5 Log findings and notify supervisor

Plan 1: if equipment is correctly examined,2-3, else 2-3-5
Plan 2: if anomalies are detected 3-4-5, else retry 1-2-3
Plan 4: if faults can be resolved on site, 4-5, else 2-3-4

Video on Explanation of HTA:
https://youtu.be/9S5uY1QXfA8

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HIERARCHICAL TASK 3 Plan: Following Structured Step
0 Following structured steps
    1 Access step-by-step procedure
    2 Review safely requirements for the tasks
    3 Execute each step while monitoring guidance of procedure
    4 Verify task completion and compliance with safety standards
    5 Verify task completion and compliance with safety standards

Plan 4 : If task is being executed, 4-5, else 2-3-4
Plan 5 : If there are unexpected issues during execution 4-5, else retry 3-4-5

Video on Explanation of HTA:

https://youtu.be/G2wzcW5nyIk

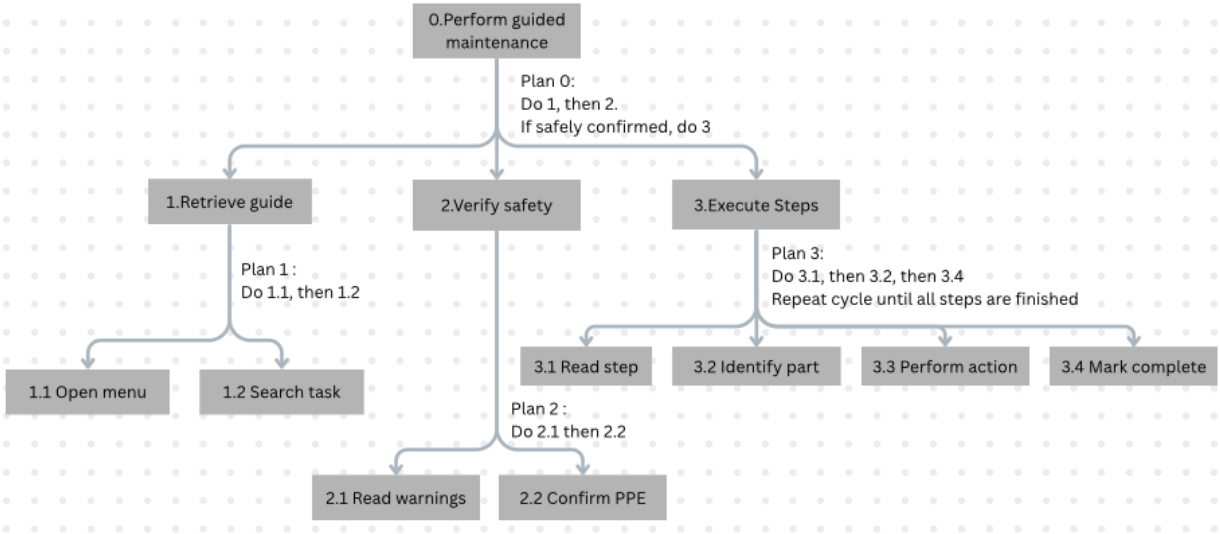
Based on Persona 2, who is a supervisor at an oil rig company :
HIERARCHICAL TASK 1 Plan: Monitor & Interpret Data

0.Monitor & interpret data
- 1. Access system
  - 1.1 Login
  - 1.2 Navigate
- 2 Locate equipment
  - 2.1 Scan layout
  - 2.2 Search ID
- 3 Interpret matrics
  - 3.1 Read value
  - 3.2 Click help
  - 3.3 Compare
- 4 Report status
  - 4.1 Flag as normal
  - 4.2 Flag is issue

Plan 0: Do 1, then 2, then 3. If value is safe do 4.1. If value is unsafe do 4.2
Plan 1: Do 1.1 then 1.2
Plan 2: Do 2.1, if equipment not found, do 2.2
Plan 3: Do 3.1, then optional 3.2, then 3.3
Plan 4: Do 4.1 then 4.2

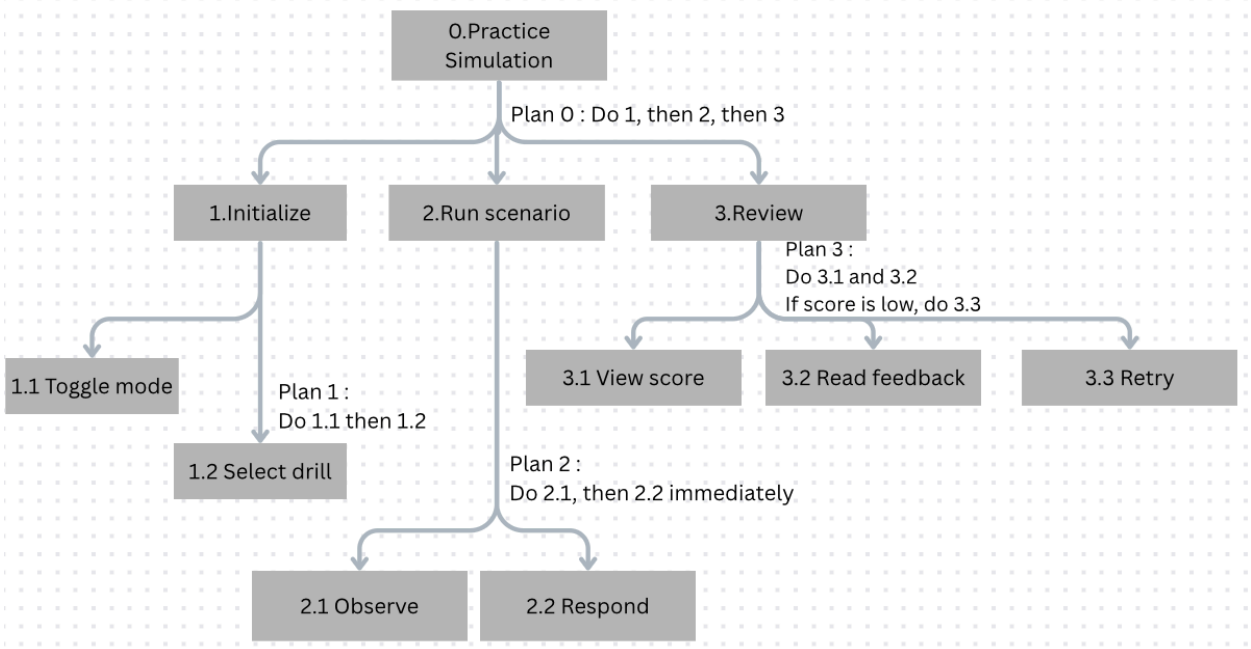
Video on Explanation of HTA:
https://youtu.be/Go2s2rf7n5I

HIERARCHICAL TASK 2 Plan : Perform Guided Maintenance



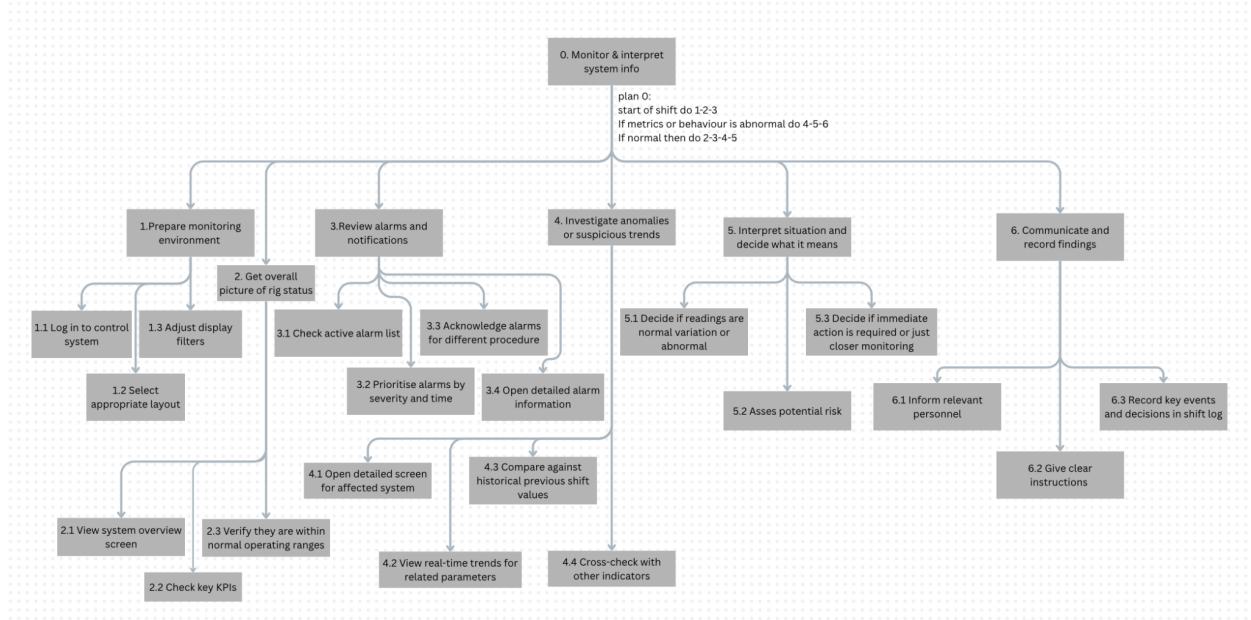
Video on Explanation of HTA:
https://youtu.be/sghVaXncS9Y

HIERARCHICAL TASK 3 Plan: Practice Simulation



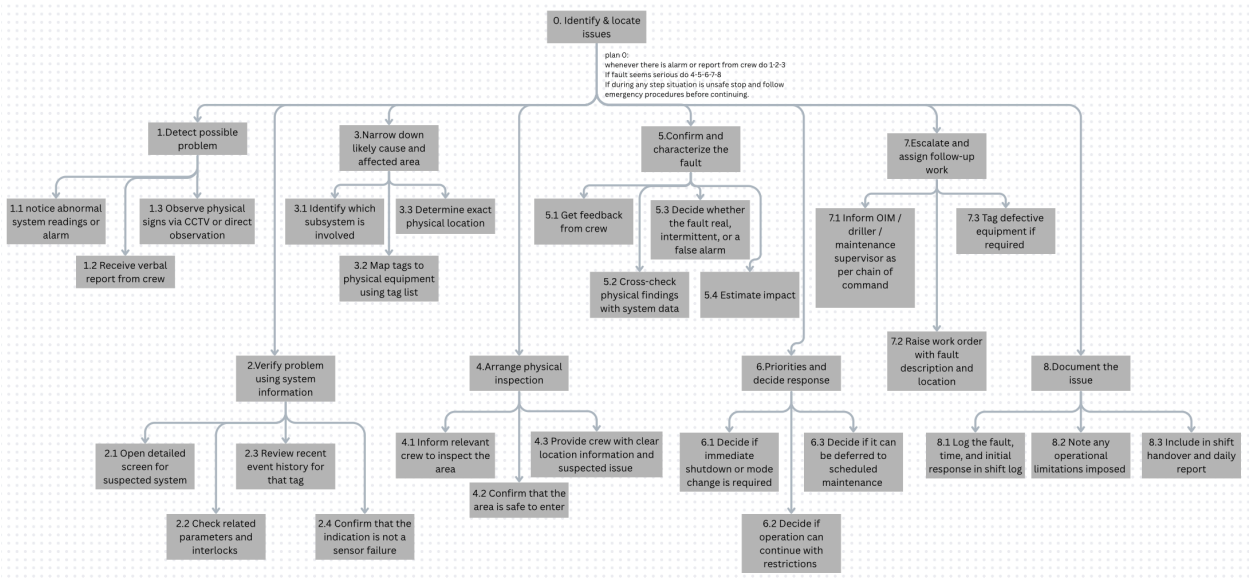
Video on Explanation of HTA:
https://youtu.be/5ea4MqgiFp0?si=ptgvBUSNzO3hQR_K

Based on the Persona 3, who is a trainee in an oil rig company :
HIERARCHICAL TASK 1 Plan: Monitor & Interpret System Information



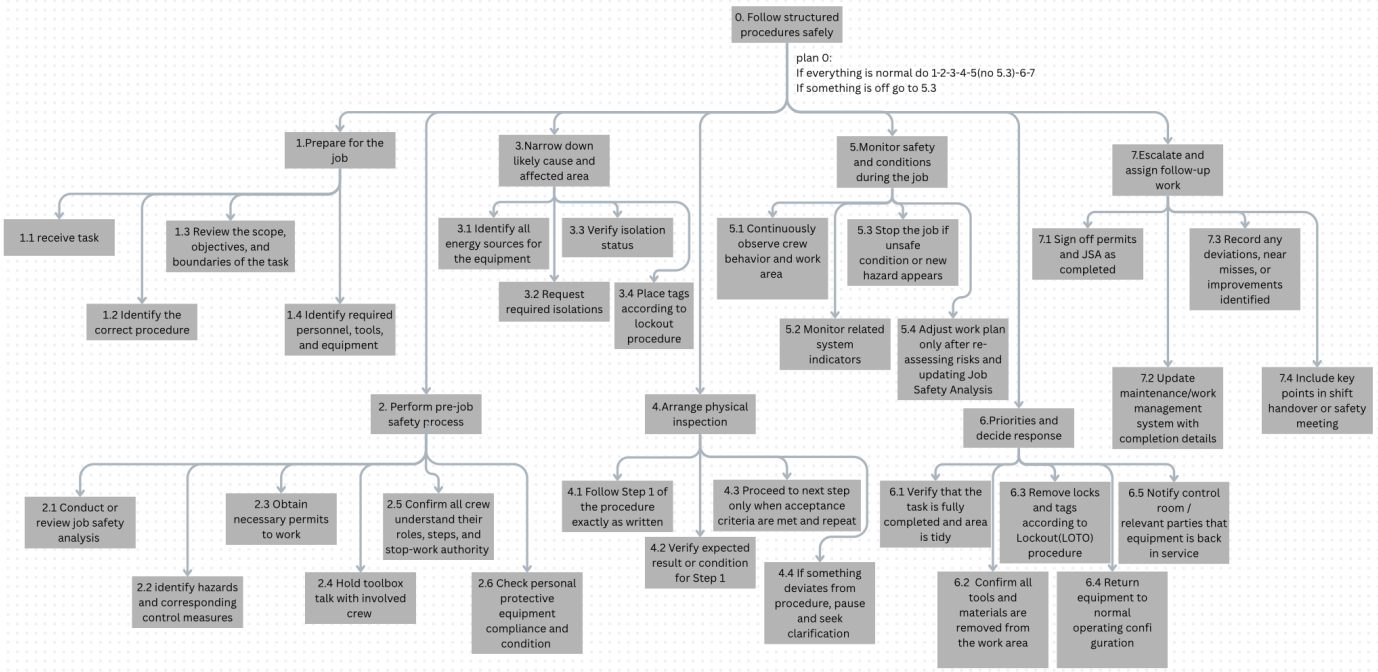
Video on Explanation of HTA:
https://youtu.be/1Gcbhn1ng0Q?si=pN85PEBwx3QrF8jY

HIERARCHICAL TASK 2 Plan : Identify & Locate Issues



Video on Explanation of HTA: https://youtu.be/ffnUViVW4zU?si=dSwoIkcbDoXpU9-J

HIERARCHICAL TASK 3 Plan : Follow Structured Procedures Safely



Video on Explanation of HTA:
https://youtu.be/MJ9muR21Xnw?si=sHxJuMc44P8YsL0j

DISCUSSION

Findings and Differences Between Users
User group 1 : Maintenance Engineer
Task 1 : Monitor Equipment or System Information
Task 2 : Identify or Locate Issues on Equipment or Around the Rig
Task 3 : Follow Structured Steps And Procedures to Perform the Tasks Safely

Key Observation :

- Usability Issues :
  Most of the time, engineers work by inspecting equipment and monitoring data. However, it is difficult for them to interact with the digital tools as they are wearing safety gloves and working in noisy environments. This situation results in delays when accessing critical data.

- Insights :
  A hands-free solution would help in improving the task efficiency and reducing the safety risks. For example, the AR Smart Helmet will provide real-time data directly to the engineer's line of sight, which will allow them to still focus on their physical inspection while monitoring the equipment.

User group 2 : Supervisor
Task 1 : Monitor Equipment or System Information
Task 2 : Identify or Locate Issues on Equipment or Around the Rig
Task 3 : Follow Structured Steps And Procedures to Perform the Tasks Safely

Key Observation :

- Usability issues :
  Supervisors face challenges like information overload since there are too many alarms and data points on the system. Dashboard is full with non-essential data which makes it hard for the supervisor to identify the critical issues quickly. The system does not show visual hierarchy for critical alerts.

- Insights :
  A role specific dashboard which prioritises alarms and also highlights critical information will significantly help in improving the supervisor's ability to make quick decisions. By having color coded alerts and clear visual indicators, it will help in guiding supervisors directly to the most critical issues in that moment.

User group 3 : Trainee
Task 1 : Monitor Equipment or System Information
Task 2 : Follow Structured Steps And Procedures to Perform the Tasks Safely
Task 3 : Practice Operating Machinery and Tools Under Supervision

Key Observation :

- Usability issues :
  Trainees usually have difficulties with the complex system data and need guidance for them to interpret the information correctly. The system's terms might be too overwhelming for the trainees to learn. The interface can be too complex and the guide of the troubleshooting lacks clarity for the beginners.

- Insights :
  Step by step tutorials and visual aids will help in simplifying the learning process for the trainees. When the trainees use the context sensitive help, it will assist them in following the procedures more efficiently without a wrong step. By using the AR Smart Helmet, it can overlay real-time guidance, which will help the trainees to do the task correctly without the need for them to ask the others for guidance.

Insights from Think-Aloud Sessions

1. Hands-free data access helps maintenance engineers to focus on the physical tasks while they are still monitoring the systems.

2. A clear and prioritised alert is needed for the supervisors to quickly respond to the critical issues without getting distracted by the irrelevant information.

3. Guided interactions such as visual feedback and step by step instructions can help train the new trainees to learn without stressing over wrong steps. They also get to improve their learning process.

Lessons Learned

1. **Simplified Interfaces for Non-Technical Users**
   Non-technical users such as supervisors and trainees can benefit from the simplified dashboards which focus on the critical data. As for supervisors, they might need role-specific dashboards to prioritise critical alarms, while as for trainees, need clear and easy to follow instructions to make the learning process become easier. By simplifying

the interfaces, it helps in reducing the cognitive overload and making sure that the users can make decisions quickly without distracting to the unnecessary data.

2. **Hands-Free Interaction for Engineers**
   The efficiency and safety of an engineer in the oil rig can be improved when there is access to data by hands-free. The need for continuous data review without having to stop on their ongoing tasks is important for the engineers to improve the efficiency. Disruptions should be minimised to their workflow to help them stay focused on their physical tasks and at the same time staying informed about the system performance.

3. **Real Time Feedback for Trainees**
   Trainees feel overwhelmed by the complex tasks and also unclear guidance for the system in the oil rig. By providing real-time feedback and structured guidance help the trainees to build confidence while they are learning about the system, and the task accuracy also gets to be improved. When there is support to ensure that the task is executed correctly, the process of learning can get smoother.

4. **Role-Based Personalisation Enhances Decision Making**
   Supervisors need focused data that highlight the most crucial information to ensure swift responses. Tailoring the dashboard can help users to navigate the system more effectively, and lead them to improve the response times. They can also ensure the task completion within the time given.

DESIGN REQUIREMENT

1. **Role-Based Dashboards**
   Dashboards should be able to personalise based on the user's role. Supervisors will need a simplified view that shows the critical alarms. Maintenance engineers on the other hand require detailed system diagnostics, while trainees need a guided instructions and visual aids to help them learn the systems.

2. **Hands-Free Interaction for Engineers**
   When working in noisy and physically demanding environments like at the oil rig, the workers will prefer to have hands-free access to real-time data. Engineers can monitor the machinery without interrupting their physical tasks. Easy ways of interaction with the system such as voice commands will make it convenience for the maintenance engineers to handle.

3. **Simplified Guidance for Trainees**
   When performing the tasks, trainees need clear and step by step guidance. The AR Smart Helmet should overlay interactive tutorials and context-sensitive help which will directly

be in their line of sight, simplifying the complex tasks and improving their learning confidence.

4. **Real-Time Data and Alerts**
   The system should show the real-time data and critical alerts in an easy to interpret format. By using the AR Smart Helmet, the users can stay informed by displaying the most crucial data in their field of vision. Users also get to reduce the distractions and enable a quick decision making process.